

Chapter 12

More Realistic and Complex Pricing

Learning Objectives

- 12.1 Understand the process of pricing commonly owned substitutes.
- 12.2 Understand the process of pricing commonly owned complements.
- 12.3 Understand the process of setting price to fill available capacity.
- 12.4 Infer how forecast of demands influence pricing.
- 12.5 Interpret the role of promotional expenditure on demand and thereby, the price of a product.
- 12.6 Investigate the role of psychological biases in formulating pricing strategy.

Chapter 12 – Main Points

- After acquiring a substitute product,
 - raise price on both products to eliminate price competition between them.
 - raise price more on the low-margin (more price elastic demand) product.
 - reposition the products so that there is less substitutability between them.
- After acquiring a complementary product, reduce price on both products to increase demand for both products.
- If fixed costs are large relative to marginal costs, capacity is fixed, and $MR > MC$ at capacity, then set price to fill available capacity.
- If demand is unknown, and the costs of underpricing are smaller than the costs of over-pricing, then underprice, on average, and *vice-versa*.
- If promotional expenditures make demand more elastic, then reduce price when you promote the product, and *vice-versa*.
- Psychological biases suggests “framing” price changes as gains rather than as losses.

Warm-Up

- What's the best deal you ever got, and what made it feel like a 'win'?

Commonly Owned Substitutes

- Let's consider a fast-food restaurant, Tasty Fries, that sells sweet potato fries.
- It acquires a competitor, who sells regular fries.
- If Tasty Fries lowers the price of sweet potato fries, they risk losing sales on regular fries (so called **cannabilization**)
 - What should Tasty Fries do?
- Tasty Fries should consider both products together as a portfolio or “bundle” of goods
 - Demand for a bundle of substitutes is less elastic than demand for the individual products - less elastic demand implies a higher optimal price (MR may be $< MC$)
 - Raise price on both goods, but raise the price more on the lower margin product (try to push price-sensitive customers to the higher-margin product)

Commonly Owned Substitutes

- Tasty Fries wants to avoid cannibalization and increase the profit of their fries. They choose to increase the price on both goods. We also know they will increase the price on the lower margin product
- Suppose they have the following prices and marginal costs:
- $P_S = \$3$, $MC_S = \$1.5$, $P_R = \$2$, and $MC_R = \$1.50$ such that they get the following margins $\%((P-MC)/P)$:
$$(\$3 - \$1.5)/\$3 = 50\% \quad \text{and} \quad (\$2 - \$1.5)/\$2 = 25\%$$
- This makes the regular fries the lower margin good, and the one Tasty Fries should increase price by more.
- One last way that Tasty Fries can avoid cannibalization is to market the sweet potato fries in a way that distinguishes them such as a “healthy fry”

Commonly Owned Complements

- Again, consider the two products a bundle, as they will be consumed as such
- If Tasty Fries acquires Happy Hot Dogs, then we will expect consumers to purchase these goods together.
- Before acquisition, 100 hot dogs and 100 fries were sold at \$3 and \$2 each for combined sales of \$500.
- After acquisition, 150 hot dogs and 150 fries were sold as a bundle at \$4 for combined sales \$600.
- The implication behind this is:
 - Tasty Fries earns more in sales causing their marginal revenue to increase
 - Now that $MR > MC$, we have a more elastic demand for both goods
 - To restore the profit-maximizing level of sales, price should be lowered on both goods

Commonly Owned Complements/Sub.: Pricing Strategy

When you own multiple substitute products, you want to minimize cannibalization and maximize total profit, not just volume.

Relationship	Raise or Lower Price?	Pricing Goal
Substitutes	Raise both (more on elastic good)	Minimize cannibalization
Complements	Lower both	Boost joint sales and MR

Question	Strategy
Are products close substitutes?	Avoid pricing them too similarly
Which product has higher margin?	Push demand toward it via lower price
Is elasticity higher for one product?	Raise price more on more elastic one
Is one product brand-preferred?	Price the preferred one higher

Key Idea: For substitutes, diverge prices to steer customers toward the product that gives you more profit per unit.

Activity – Price Bundling Trade-offs

Scenario:

- You're managing pricing for two products your company owns:
 - Product A: A digital textbook
 - Product B: A subscription study guide
- Product A and B can be sold separately or bundled. You also own a competing textbook (Product C). How you price each will affect demand across the others.

Questions:

1. How would you price the textbook and study guide if they are complements?
2. Would you bundle them? Why or why not?
3. How would your strategy change if the second textbook (Product C) is a substitute for Product A?
4. What pricing risks do you see with cannibalization or demand confusion?

Activity – Price Bundling Trade-offs

1. If A and B are complements:
 - Price A and B slightly below standalone profit-maximizing price to boost bundle volume.
 - Consider offering a bundle discount (e.g., \$50 for both vs. \$30 each).
2. Bundle?
 - Yes, if customers typically buy both — bundling simplifies decision-making and reduces price sensitivity.
 - No, if customer overlap is low or if flexibility is valued.
3. If Product C is a substitute for A:
 - Lower the price of A slightly or differentiate it (features, brand).
 - Avoid cannibalizing C unless you're trying to phase it out or reposition it.
4. Pricing Risks:
 - Cannibalization: If C pulls sales from A.
 - Confusion: If pricing isn't clear, students may delay or skip purchasing.
 - Over-discounting: Bundling may lower perceived value if not positioned carefully.

Discussion Activity 1

- Why might you *raise* the price of a low-margin product in a bundle with a high-margin product?
- Think about cannibalization, elasticity, and product repositioning. Does pricing steer customers?

Revenue or Yield Management

- Firms often are faced with capacity constraints: hotels, ballparks, arenas, airlines etc.
- First, begin by making the distinction between long run and short run
 - These particular firms will often have very large upfront fixed (sunk) costs and low marginal costs
- In the long run, **build capacity** until $LRMR = LRMC$
 - Since firms incur the sunk cost, they should only consider adding more seats, rooms, etc. until the above holds
- In the short run, **price to fill capacity**: $SRMR > SRMC$ (or $MR > MC$)
 - Ignore fixed costs, as those have already been incurred
- How do ballparks handle this issue?
 - They often sell one and three game packages at lower prices
 - They also keep select seats available for high value customers on game days

Revenue Management

- Designing a new hotel:
 - Keep adding rooms to the design plan, as long as $LRMR > LRMC$
- Suppose that the optimal size is 300 rooms. At the optimal size, annualized LRMC of building, cleaning, and heating the room is about \$400 per day.
- Once the rooms are built, the costs have been sunk, the hotel's owners must decide what to charge for the rooms. Suppose that 90% of the annualized LRMC are fixed or sunk and that the relevant marginal cost is just \$40 per day.
- Since capacity decisions are determined by all costs, and the pricing decision only by short-run MC, it's likely that $MR > MC$ at the capacity of the hotel. If so, then the hotel's owner should price to fill capacity (sell all available rooms).
- Choose a price that matches *expected* demand to capacity
 - In some industries, like parking lots, stable demand and the daily observation of realized demand make this relatively easy to do
 - In other industries, like cruise ships, this is much more difficult

Long Run Analysis – Stadium Expansion

- Current capacity = 50,000 seats
- Sells out 25 times per season
- Average ticket price = \$100
- Proposal to add 10,000 seats
- Estimated sell rate of new seats is 75%
- Revenue from new seats: $10,000 \times 0.75 \times 25 = \$75,000$
- Construction costs = \$1 million over 5 years or \$200,000/year
- Additional staffing and maintenance costs = \$20,000 per year
- LRMR (\$187,500) < LRMC (\$220,000)
- Don't expand
 - Stadium would need to expand to 11,734 seats for LRMR = LRMC

Discussion Activity 2

- Airlines and hotels often charge wildly different prices for similar products. Why?
- Think about fixed vs. marginal cost, capacity, and consumer willingness to pay at different times.

Forecasting Demand and Price

When demand is uncertain, pricing decisions must account for risk, inventory constraints, and the cost of getting it wrong.

Dilemma:

- Underprice – ensures high volume and avoids unsold stock, but leaves money on the table
- Overprice – higher profit per unit (assuming demand materializes), but lost sales and inventory risk

Forecast Certainty	Cost of Underpricing	Cost of Overpricing	Recommended Strategy
High uncertainty	Low	High	Underprice conservatively
Low uncertainty	High	Low	Price aggressively (closer to value)
Balanced forecast	Medium	Medium	Choose middle-ground pricing
Known scarcity	Very low	Low	Price high to ration demand

Forecasting Demand and Price

- Example: You're launching a limited-edition product with unknown popularity.
 - If you underprice: sell out fast, no leftovers
 - If you overprice: stuck with inventory or price backlash
- If forecast is uncertain, price slightly below expected willingness to pay, then adjust in future batches.

Use Data to Improve Forecasting

- Past sales patterns
- Preorders or waitlists
- Social media mentions
- Elasticity estimates

Forecasting Demand with Uncertainty

- Cruise ship has 2,000 rooms for a 6-day cruise.
- They create two average prices: \$2,000 and \$3,000.
- At the lower price, they have an 80% chance of booking 90% of the rooms and a 20% chance of booking all rooms.
- At the higher price, they have a 60% chance of booking 70% of the rooms and a 40% chance of booking 50% of the rooms.
- Low price: $(0.8 \times 1800 \times 2000) + (0.2 \times 2000 \times 2000) = \3.68M
- High price: $(0.6 \times 1400 \times 3000) + (0.4 \times 1000 \times 3000) = \3.75M
- Charge the higher price

Activity – Demand Forecasting Under Uncertainty

Scenario:

- You're advising a cruise line preparing for holiday bookings. They must choose a pricing strategy before knowing exact demand. They forecast:
 - High demand (70% chance): \$1,000 tickets sell out (100 rooms)
 - Low demand (30% chance): \$1,000 tickets only sell 40 rooms
- Alternative: Price at \$800 to sell 80 rooms regardless of demand.

Questions:

1. What is the expected revenue from each pricing option?
2. Which pricing strategy would you recommend based on expected value?
3. What other factors (capacity, brand image, risk) might affect your advice?

Activity – Demand Forecasting Under Uncertainty

1. Expected Revenues:

- Option 1 – \$1,000 Price:
 - 70% chance of selling out ($100 \times \$1,000 = \$100,000$)
 - 30% chance of selling 40 tickets ($40 \times \$1,000 = \$40,000$)
 - Expected Revenue = $(0.7 \times 100,000) + (0.3 \times 40,000) = \$82,000$
- Option 2 – \$800 Price:
 - Always sells 80 tickets: $80 \times \$800 = \$64,000$

2. Recommendation:

- Choose \$1,000 strategy if goal is to maximize expected revenue.
- Choose \$800 if risk aversion, capacity planning, or brand strategy favors consistency.

3. Additional Factors:

- Brand image: Higher price may reinforce exclusivity.
- Capacity constraints: If fixed costs are high, underutilization may be costly.
- Reputation risk: Empty rooms could signal poor demand.

Advertising and Promotional Pricing

- Marginal analysis can help us determine what to do with product price once advertising occurs.
 - One goal of advertising is to make your product more inelastic (less elastic) so that consumers will be less responsive to price changes
 - If the above holds, and the demand for the good becomes less elastic then the firm should raise the price of the good.
 - If on the other hand, your promotion (because it focuses in on the attractiveness of the good) makes consumers more inelastic, then you want to raise the price of your good.

Psychological Pricing

- What happens if the consumer doesn't behave rationally as the rational-actor paradigm predicts? Why does a consumer value an \$8 hot dog at a ballpark but not in the grocery store?
- Behavioral economics has been looking at these types of questions to better understand why consumers seem to act irrationally.
- Prices may be based on a distance from a reference price (position).
 - That is, a consumer may consider their environment or other event as to why they are spending more for an item.
- So, a firm should try and make the customer perceive a win with a price (position) below their reference price (position)
- The key behind it is that people prefer gaining less to losing more, *i.e.* given a two scenarios where 1) you lose \$10,000 or gain \$20,000 and 2) you lose \$10 or gain \$20 on a coin flip, you will, most likely, agree to 2) but not 1).

Psychological Pricing

- Consider a firm selling three types of cars: a base model that costs \$15,000, and two other versions that have upgraded options that cost \$20,000 and \$25,000.
- For the consumer, under prospect theory, they feel the loss of \$5,000 (going from the second model to the base model) more than they perceive the \$5,000 in gains from going from the second model to the third model.
- What to do?
 - Promote trials, so the perceived loss in utility is effectively 0 if the customer returns the product
 - Promote good-better-best lines to consumers and move them through the chain or to closely related products.

Psychological Pricing Summary

Strategy	Description
Reference pricing	Create a high fake “original” price
Framing	Frame price drops as gains, not losses
Decoy pricing	Use middle options to steer choice
Trial/promotion	Reduce perceived loss risk
Prestige pricing	Use high price to signal quality

Case Study: JCPenney's "Fair" Pricing Failure

- Richard Thaler contends there are two types of utility: *acquisition utility* (equivalent to consumer surplus—i.e., the gain in utility after subtracting our opportunity cost) and *transaction utility* (the difference in the price paid and a reference price)
- When we purchase an item, we get both types of utility. Sales present deals that may be too good to pass up, hence increasing our transaction utility.
- What can a seller do? Offer a large “fake” suggested retail price. This establishes a high reference price.
- If a retailer offers continuous sales or discounts, then weaning their consumer base off them may cause backlash. [Enter JCPenney](#).
- Former CEO Ron Johnson did away with “fake” prices and created a simpler pricing scheme such as rounding the .99 up to the nearest dollar.
- The new pricing was practically the same as the old pricing but was a huge failure.
- Why? Because consumers were losing out on their transaction utility.
- JCPenney stock price plummeted. Johnson was fired. And JCP returned to their original pricing and sales.

Discussion Activity 3

- Why did JCPenney's "fair pricing" strategy fail even though prices didn't change much?
- Think about transaction utility and what customers value in a pricing experience.

Outlet Malls and Psychological Pricing

- [About 55 million Americans](#) shop each year in over 300 outlet malls.
- Total trips by Americans totals enough times to circumvent the earth 440,000 times.
- Why are we going to these outlet malls? The great prices of course?
- Why are we so willing to travel so far to get these low prices? Consider the trade off between convenience and price—i.e. is travelling 100 miles vs. 10 miles worth the extra 25-50% off “retail”?
- “A trip to the outlet mall is not passive, not simply a matter of popping in to pick up a few things. We have to work to get there, piling up hefty “sunk costs.” All that time! All that gas! “I gave up my entire Sunday afternoon and even missed the game to come here!” Psychologically speaking, all this and more must be repaid in the form of purchases made.”
- An obvious, though it might not be, tactic is for stores to use psychological pricing by listing a suggested retail price, followed by a severely marked down price to alter a consumer’s willingness to pay.
- One example is a diamond encrusted pendant with a suggested retail price of \$3,329. It was marked down to \$832.
- It was later found out that similar pendants were being sold on eBay for \$299.

Other Types of Pricing

- **Prestige pricing** – setting the price of your product well above the market price for similar items.
 - e.g., Mercedes (after discounting the added quality of amenities)
- **Price lining** – the practice of setting a price target then develop a product to maximize profits at that price. Changing an established price is not likely to work.
 - Cars offer a good view of this. Most car makers have different tiers of quality and moving a car from one tier of pricing to the next would not work well with consumers.
- **Skimming** – the practice of setting an initial high price then gradually lowering the price
 - This occurs when firms don't know the strength of demand for a product. Durable goods are culprits of this. (Could be technology and/or cost changes too)
- **Value pricing** – the selling of goods at much lower prices than previously
 - e.g., Value menus at McDonald's, Burger King, Taco Bell, etc.
- **Price matching** – the practice of matching a competitor's price.
 - This allows firms to set initial high prices and lower prices to “match” a competitor.
 - This practice may be dubious, as it lowers the incentive for a consumer to search for a better price.

Pricing Strategy Summary

Strategy	Use When...	Goal	Example
Price Low-Margin Substitutes Higher	You own multiple substitute products with different margins	Steer customers to high-margin items	House vs. premium wine
Price Complements Lower	You sell items that are used together (joint demand)	Stimulate bundle sales, increase total profit	Printer + ink, burger + fries
Price to Fill Capacity	You face fixed, limited capacity and predictable demand	Maximize revenue per unit of capacity	Hotel rooms, airline seats
Forecast-Driven Pricing	You can predict changes in demand or cost before the market	Anticipate peak demand, manage inventory	Ride-sharing surge, concert tickets
Promotion to Shift Demand Curve	You want to increase awareness or attractiveness (especially if elasticity is high)	Increase quantity sold or willingness to pay	Coupons, advertising campaigns
Psychological Pricing	Buyers are sensitive to perceived fairness, reference prices, or "deals"	Tap into biases, increase conversion	Outlet malls, \$0.99 pricing

Key Takeaways

- When managing commonly owned substitutes, avoid cannibalization:
 - Raise prices on both, especially the more price-sensitive item
 - Reposition products to reduce direct competition
- When managing complements, lower prices on both to increase joint demand.
- Revenue management:
 - Long run → build capacity until $LRMR = LRMC$
 - Short run → price to fill available capacity if $MR > MC$
- When demand is uncertain:
 - If overpricing is riskier → underprice
 - Use data (preorders, past sales) to improve forecasts
- Promotions affect elasticity:
 - If demand becomes more elastic → lower price during promotion
 - If less elastic → raise price to capture more surplus
- Psychological pricing matters:
 - Reference points, perceived gains/losses, and framing affect willingness to pay
 - Avoid eliminating customer-perceived value even if price logic seems rational

Knowledge Check

1. What is the goal of pricing commonly owned substitutes?
2. What condition supports pricing to fill capacity?
3. How should a firm price under high demand uncertainty and high cost of overpricing?
4. What happens if advertising makes demand more elastic?
5. Why do outlet malls use “list price slashing” as a strategy?

Knowledge Check Answers

1. Minimize cannibalization; raise prices to reduce internal competition
2. When $MR > MC$ and fixed costs are already sunk
3. Underprice — better to sell out than hold excess stock
4. Lower the price during promotion to take advantage of increased sensitivity
5. To increase transaction utility and justify sunk costs (e.g., travel, time)

Consultant's Challenge: The Pricing Puzzle

Scenario:

- You're consulting for a small company that just acquired a competing product and is also launching a new complementary service. They have limited marketing budget, uncertain demand forecasts, and pressure to fill unused capacity.

Your Task:

1. How would you advise them to set prices for both the substitute and complementary products?
2. Which factors: demand forecasting, promotions, or psychological pricing would you emphasize most to avoid mispricing?